ESD PROJECT REPORT

Analog Active Noise Cancellation System

Fourth Semester Electronics and Communication Engineering

Submitted by

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ABSTRACT

The project's primary aim is to design and test an analog active noise cancellation (ANC) circuit mainly for low-frequency background noise using simple and low-cost circuitry. A reference input is senses the low frequency noise, which then is amplified and conditioned. This processed signal is then phase-shifted to create anti-noise. This anti-noise is combined with the output path through a summing amplifier. Thus, when anti-noise matches the incoming noise, destructive interference occurs and the perceived noise level decreases. This project focuses on improving listening comfort.

PROJECT DETAILS

Introduction:

This circuit has three parts: a pre-amplifier a delay filter and a summing amplifier.

When you look at noise-cancelling headphones you will see a hole on the outside of each ear cup. Inside this hole is a microphone. These microphones pick up sounds from outside like the noise of an engine or a fan and send them to the circuit.

The noise that the microphones detect goes to the pre-amplifier first. This part of the circuit makes the noise signal stronger so it can be used. It is a use of an op-amp where the amount of gain depends on the resistors in the circuit.

Next the stronger noise signal goes through a delay filter. This filter makes the signal wait for a while without making it stronger or weaker. This delay is necessary because sound and electricity move at different speeds. Sound moves through the air at 340 meters per second, but electricity moves almost as fast as light. If we did not delay the signal the noise that goes straight to your ear and the noise that the circuit processes would not arrive at the time. The delay makes sure they arrive together.

Then the summing amplifier combines the noise signal that the circuit has processed with the music you want to hear. The amplifier adds the two signals together. Then flips the result. This creates a wave that is the same size as the unwanted noise but has the opposite effect, which means the noise is cancelled out when it reaches your ear.

Objectives:

• To simulate an active noise cancellation circuit using operational amplifiers.

• To build a pass filter that introduces the right amount of delay, for the noise to be cancelled out.

• To see how well the pre-amplifier delay stage and summing work together to reduce unwanted noise.

Methodology:

Block diagram:

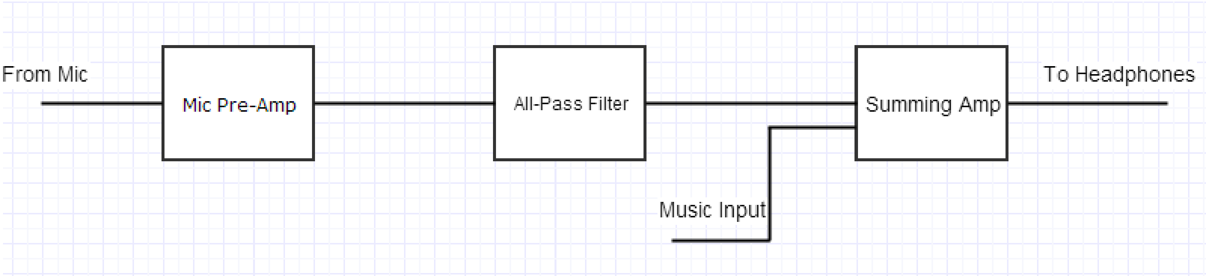

Fig 1: Block diagram

Components:

Capacitors
- 1 nF capacitor × 2
- 10 µF capacitor × 1

Resistors
- 100 Ω resistor × 1
- 1 kΩ resistor × 1
- 10 kΩ resistor × 4
- 13 kΩ resistor × 1
- 22 kΩ resistor × 1
- 235 kΩ resistor × 1
- 1 MΩ resistor × 1

Operational Amplifiers
- LT741 operational amplifier × 4

Other Equipment
- Function generator
- A handful of BNC cables and connectors
- Breadboard
- Oscilloscope
- Power supply
- Wires
- Wire strippers

Circuit diagram:

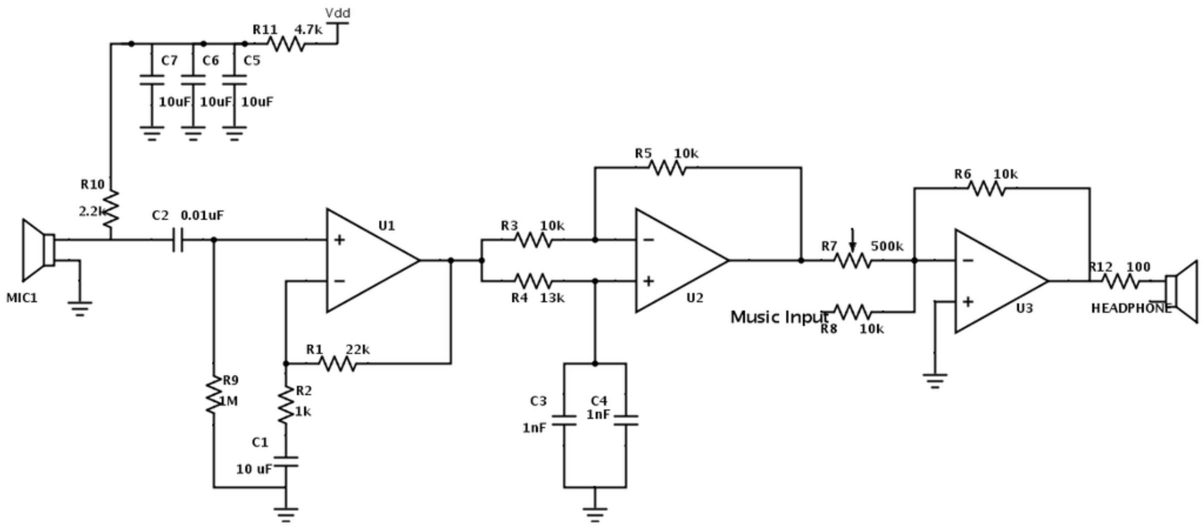

Fig 2: Noise Cancellation Circuit

Working of the system:

Power supply filter

An RC filter on V_DD reduces high-frequency supply noise. This noise could get amplified. If an electret mic is used a bias resistor gives DC bias. A coupling capacitor blocks DC. This way only the AC signal enters the pre-amplifier.

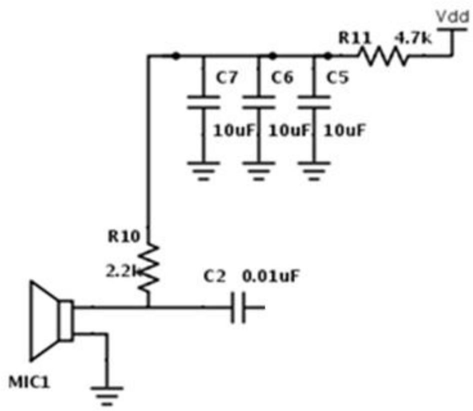

Fig 3: RC Filter

Pre-amplifier
The microphone produces a small voltage. The first op-amp stage boosts this signal to a level. The gain is mainly set by the resistor ratio. This ratio is between the feedback resistor and the input resistor. This stage helps reduce DC offsets. These offsets can build up through stages and spoil cancellation. The amplified signal then goes to the delay block.

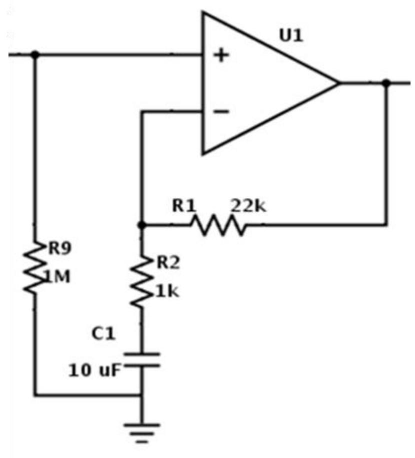

Fig 4: Pre-amplifier

All-pass filter / Delay
ANC needs correct timing. External noise reaches your ear through air at the speed of sound. The circuit processes the microphone signal faster electrically. Without correction the anti-noise could arrive early. This reduces cancellation. The pass filter adds a small delay. This delay aligns the processed noise with the noise at the ear.
The delay time is approximately equal to the distance divided by the speed of sound. The all-pass filter adds a small delay to align the processed noise with the real noise at the ear: $t \approx \frac{L}{V_s}$
and phase shift at frequency f: $\Delta\phi = 2\pi f t$
Component values (R and C) are tuned to get suitable phase shift over the target frequency range.

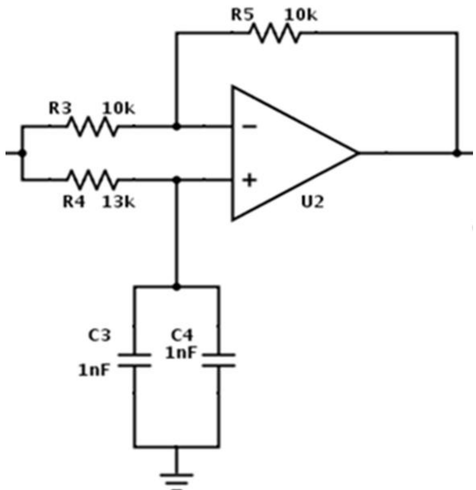

Fig 5: All-pass delay

Summing amplifier
This stage adds the music and the processed noise signal. It then inverts the result to create anti-noise. A potentiometer is often used. This adjusts the noise path level. The amplitudes must match correctly for cancellation.

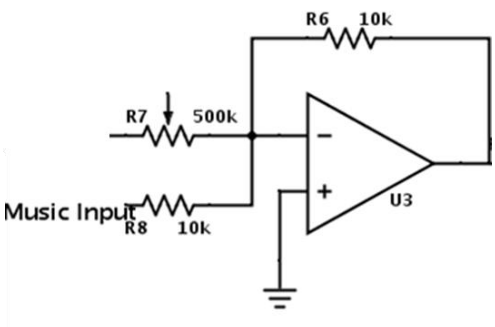

Fig 6: Summing Amplifier

Output
The final stage is a summing amplifier, on the breadboard. One input is taken from the stage which is the processed noise. The other input is the combined signal of music and anti-noise. Both signals are applied through resistors to the inverting terminal of the op-amp. This enables superposition. The resistor values are adjusted. This makes the music and anti-noise cancel each other. The output mainly contains the noise component.

Simulation Results

Schematic

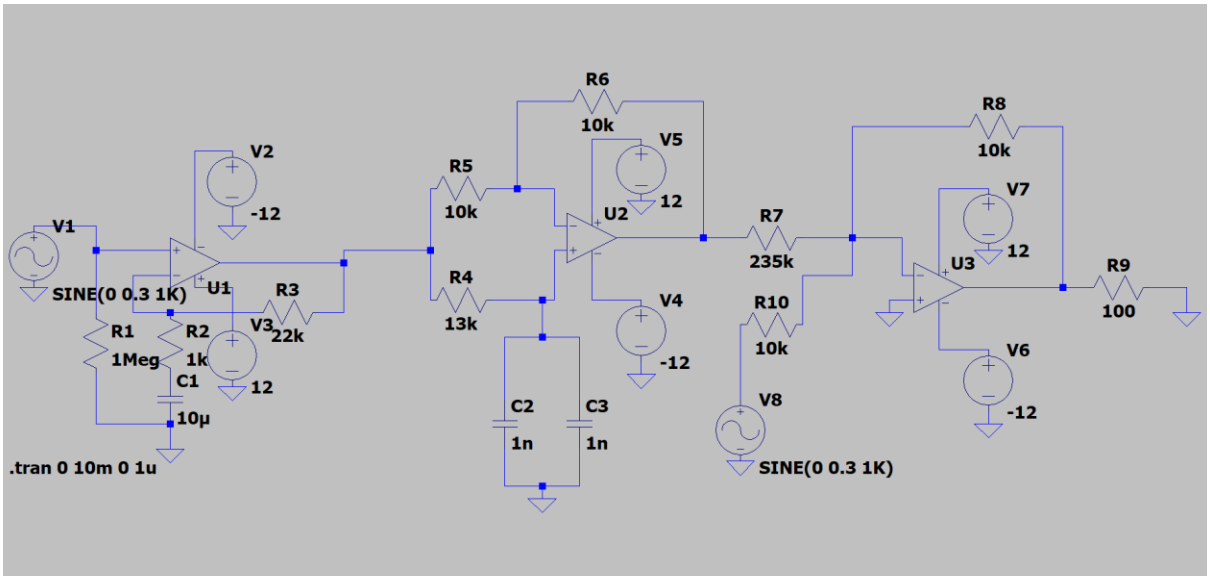

Fig 8: Schematic on LTspice

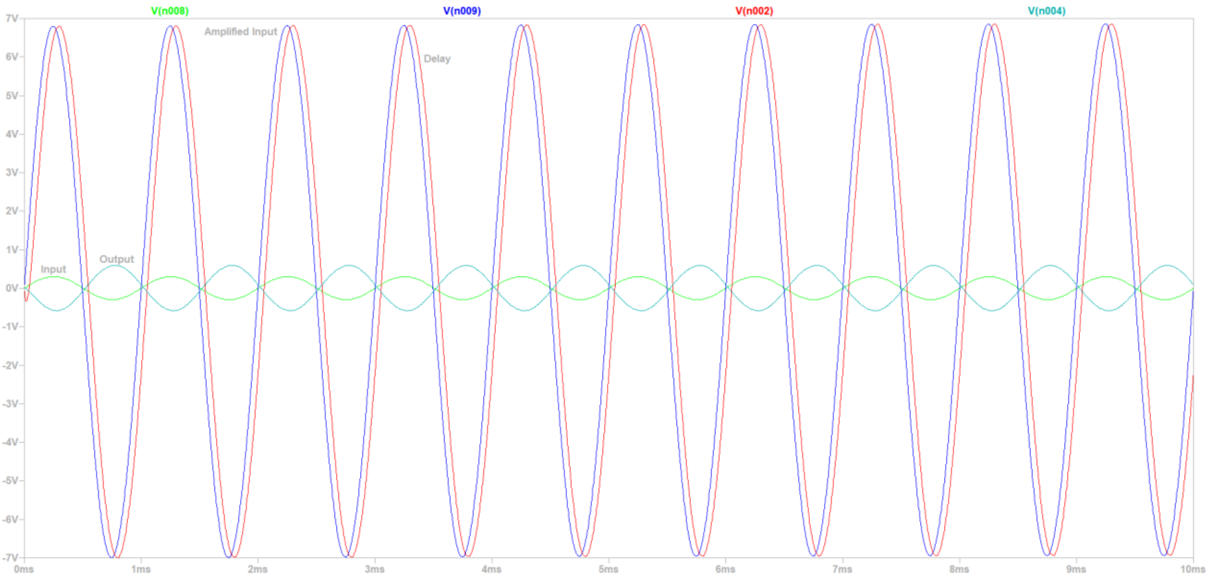

Fig 9: Simulation Results

Design Calculations

Pre-Amplifier

Gain = R3/R2 = 22k/1k = 22 times

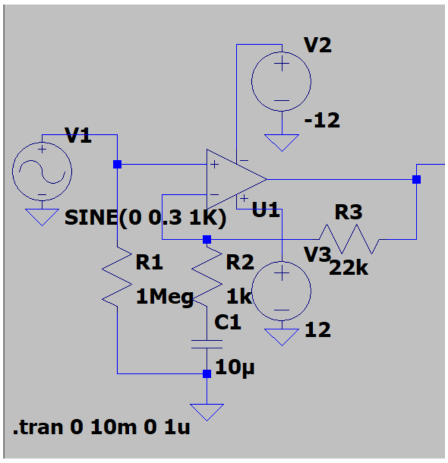

Fig 10: Pre-Amplifier on LTspice

All-pass filter/delay

$t \approx \frac{L}{V_S}$ and $\Delta\phi = 2\pi f t$

This frequency needs to be decided and is based on the range that the ANC is expected to perform. R4, C2, and C3 can be manipulated to work this out. The oscilloscope or the speaker may be very useful when working on this part.

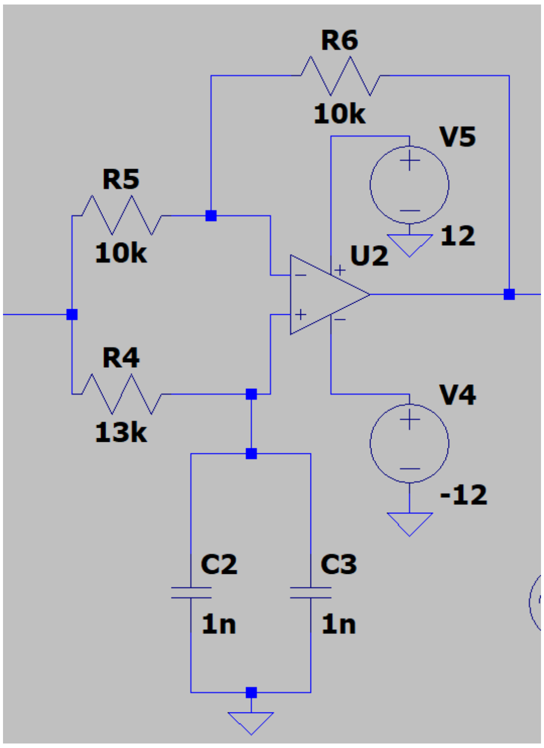

Fig 11: All-pass delay filter on LTspice

Summing Amplifier

Gain = R_8 / R_{10} = 10 kΩ / 10 kΩ = 1

Gain = R_8 / R_7 = 10 kΩ / 235 kΩ = 0.0425

These are the gain calculations for the summing amplifier. The input of the music (top) will have no gain, and the noise (below) will have a gain of 0.0425, which means that the output will be quieter.

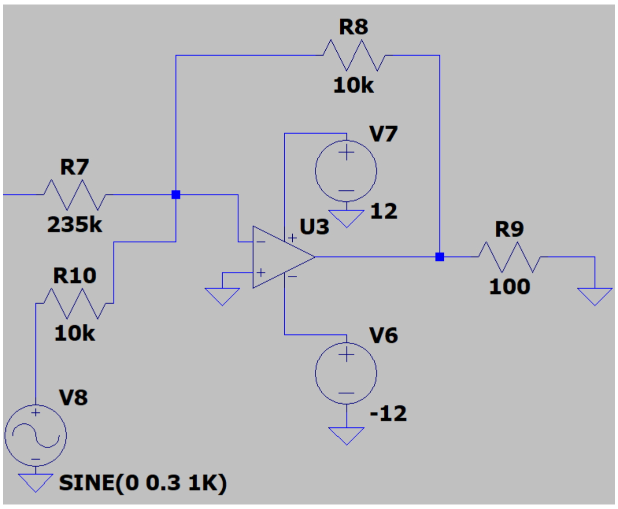

Fig 12: Summing Amplifier on LTspice

Circuit Created

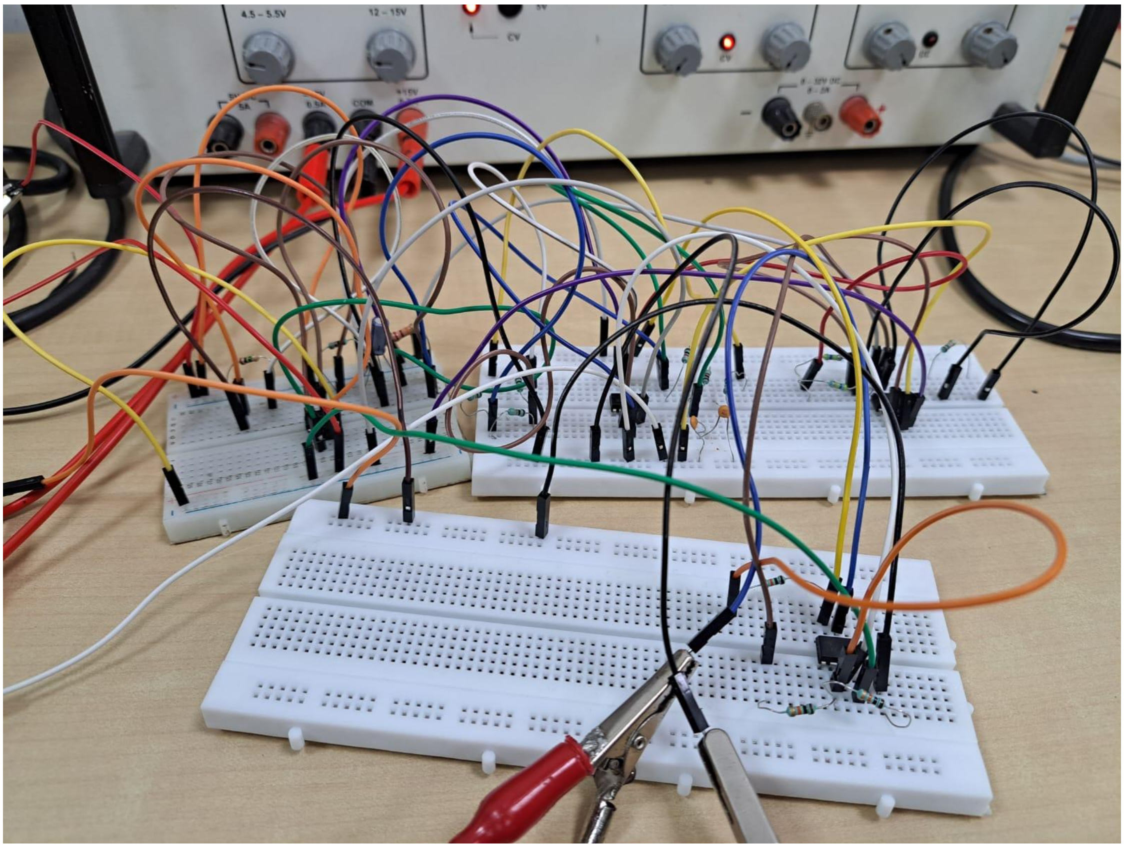

Fig 13: Final Circuit on Breadboard

Pre-Amplifier

On the breadboard, the op-amp was placed across the center gap and connected using jumper wires. Input and feedback resistors were wired to set the required gain. The noise signal was given to the input pin, and proper connections to V_DD and ground were ensured. This stage was mainly used to increase the signal level before further processing.

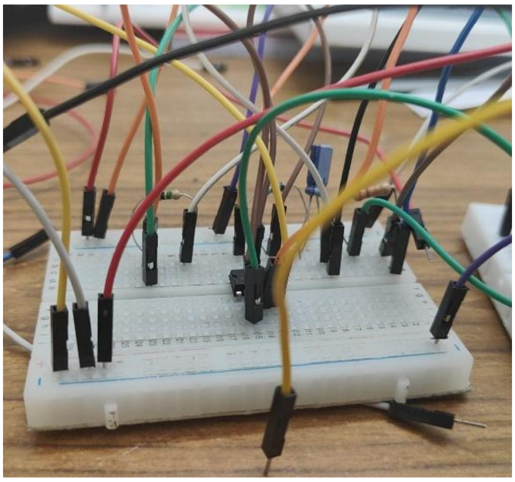

Fig 14: Pre-Amplifier on Breadboard

All-pass filter/delay

The delay stage was implemented using a combination of resistors and capacitors connected around the op-amp on the breadboard. The RC network was wired carefully to form the all-pass configuration. Component values were selected and adjusted during testing to obtain the required delay.

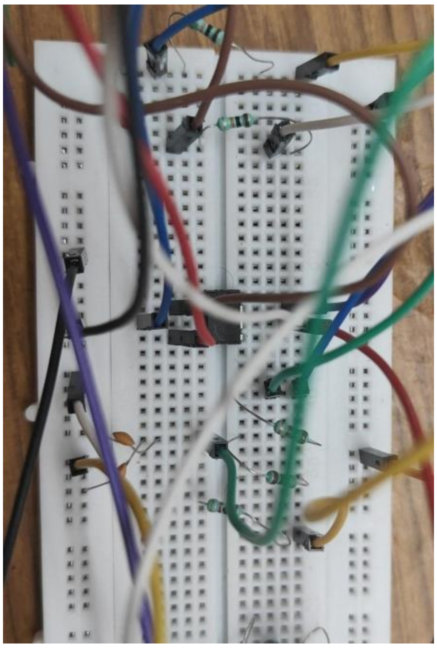

Fig 15: All-pass filter/delay on Breadboard

Summing Amplifier

For the summing stage, both the delayed noise signal and the music input were connected through separate resistors to the inverting input of the op-amp. The resistors were placed on the breadboard such that both signals meet at a common node. The output was taken from the op-amp output pin.

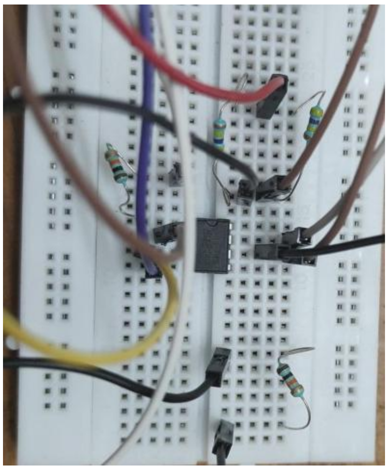

Fig 16: Summing Amplifier on Breadboard

Output

The output stage on the breadboard is implemented as another summing amplifier. One input is taken from the previous stage (processed noise signal), while the other input is the combined signal of music and anti-noise. Both signals are connected through separate resistors to the inverting input of the op-amp, ensuring proper superposition at a common node. The final output is taken from the op-amp output pin, which primarily contains the noise component.

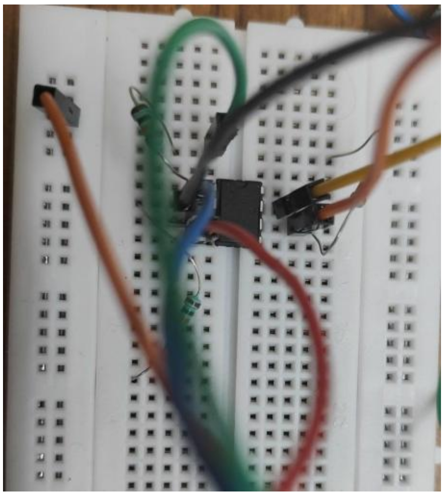

Fig 17: Output on Breadboard

Waveform

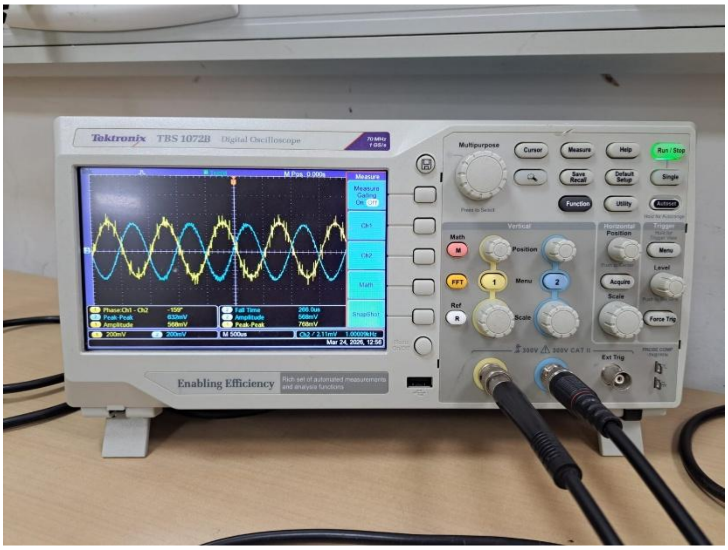

Fig 18: Inverted Noise

Applications in Real Life

- Foundation for Modern Noise Cancellation Systems:
  This circuit demonstrates the fundamental principle used in all ANC systems destructive interference. It forms the basic building block for advanced digital noise-cancelling technologies used in headphones and audio devices.

- Low-Frequency Noise Control in Controlled Environments:
  The circuit can be used to reduce predictable low-frequency noises such as fan noise, motor hum, or electrical interference in small or controlled setups.

- Analog Front-End in Hybrid ANC Systems:
  In practical systems, similar analog stages (pre-amplifier and filtering) are used before digital processing. This design represents the essential front-end required for signal conditioning.

- Acoustic and Signal Processing Studies:
  Helps in understanding real-time signal processing concepts such as amplification, phase shift, and superposition, which are critical in communication and audio engineering fields.

Future Improvements

- Use of Adaptive Filtering (Digital ANC)
  Replace the fixed analog delay with an adaptive filter (like LMS algorithm) so the system can automatically adjust to changing noise conditions and improve cancellation performance.

- Improved Delay Accuracy
  The current RC-based all-pass filter gives only approximate delay. Using multi-stage filters or digital delay lines can provide more precise phase alignment across a wider frequency range.

- Multi-Frequency Noise Cancellation
  The present circuit mainly works for low-frequency noise. Expanding the design to handle a broader frequency spectrum would make it more practical.

Conclusion

The project successfully demonstrates the basic working principle of active noise cancellation using simple analog circuitry. By implementing a pre-amplifier, all-pass filter (delay stage), and summing amplifier on a breadboard, the system was able to process the input noise signal, generate an anti-noise signal, and combine signals effectively. The circuit highlights how phase alignment and signal superposition can be used to achieve noise reduction, particularly for low-frequency sounds.

Although the performance is limited compared to modern digital ANC systems, this project provides a clear understanding of the fundamental concepts involved, such as amplification, phase shifting, and signal mixing. Overall, the design serves as a strong foundation for exploring more advanced and adaptive noise cancellation techniques in future work.

REFERENCES

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